Section 1



1. Given that it has a red-brown streak (shown on the right), identify the rock/mineral. (2)

_____

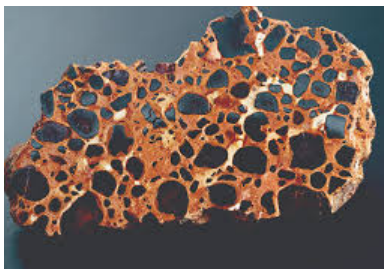
2. Answer each question below with True/False regarding the rock/mineral you identified above. (2, 0.5 each)
   a. It has a hardness of 4.0
   b. It forms radiating or tabular crystals.
   c. It does not have a cleavage.
   d. It has a specific gravity of 4.9-5.3.

3. Identify the chemical formula of the rock/mineral you identified in Question 1. (2)

_____

4. Identify the rocks below using the word bank. WORDS MAY BE USED MORE THAN ONCE OR NOT AT ALL. (9, 1 each)
Albite, Almandine. Amazonite, Apatite, Aragonite, Augite, Azurite. Bauxite, Barite

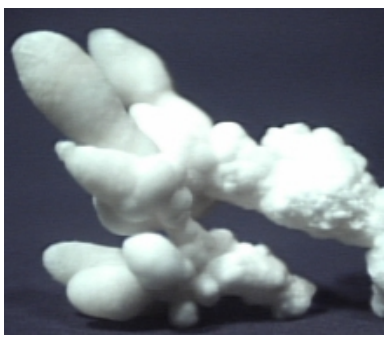
a.



b.



c.



d.



e.



f.



g.



h.



i.



Section 2

1. Areli found an igneous rock that is dark with a lot of holes and very small crystals. What can she infer from her observations about the rock's formation? (3)

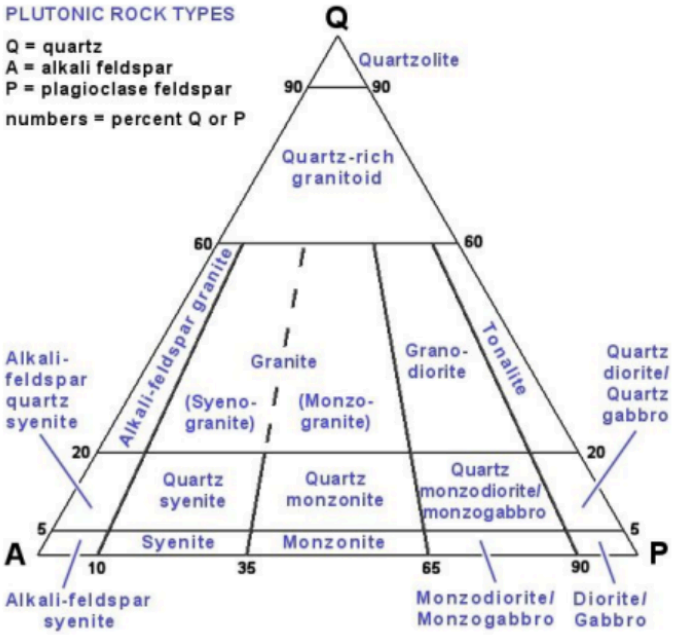
_____

_____

2. Areli went hiking and found a foliated rock that is crumbly to the touch. The rock is likely (2)
   a. sedimentary
   b. metamorphic
   c. igneous
   d. volcanic

3. Larger crystals are more likely to form _____ to the surface while smaller crystals form _____ to the surface. (2, 1 each)

4. What would a rock composed of 28% quartz and 65% plagioclase be classified as, based on this diagram? (2)



_____

5. What trace element gives aquamarine beryl its color? (2)

_____

6. Aragonite and calcite are (MULTIPLE CHOICES MAY APPLY) (3)
   a. polymorphs
   b. isomorphs
   c. isometric
   d. carbonate minerals
   e. phosphate minerals
   f. commonly formed organically

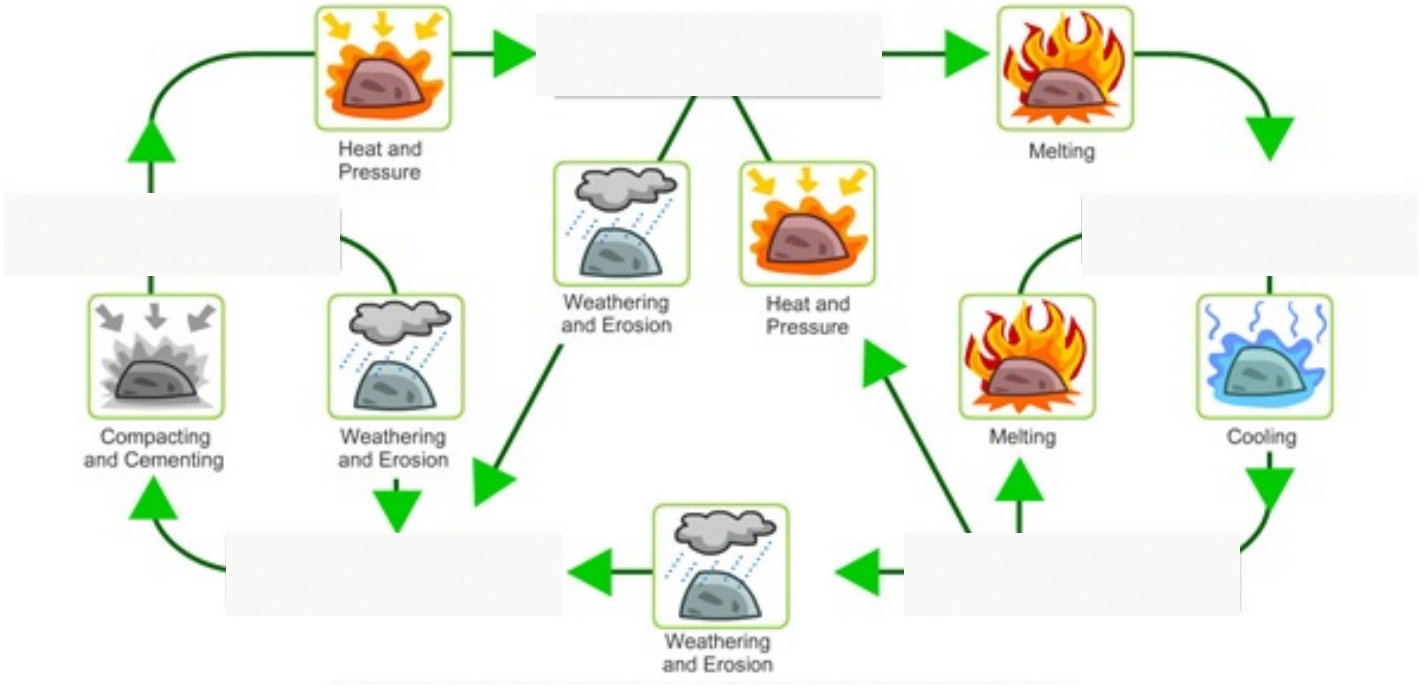
Section 3

1. Explain the difference between a laccolith and a batholith. (4)

_____

_____

2. Fill out the following chart with the words provided:
Metamorphic Rock, Igneous Rock, Sedimentary Rock, Sediment, Magma



3. Quartz is less / more susceptible to chemical weathering as compared to pyroxene. (Hint: Bowen's Reaction Series) (2)

4. What are joints in the context of rock? (4)

_____

_____

5. Use the words provided to label each of the following rocks. (5, 1 each)



Section 4

1. Identify the rock/mineral(s) in the following images. (6)

a. 

b. 

c. 

d. 

e. 

f. 

2. Describe what would happen if the rock/mineral you identified in 1a were dropped in a solution of HCl. (3)

_____

3. What is the 1d's (4, 1 each)

a. streak? _____

b. cleavage? _____

c. hardness? _____

d. specific gravity? _____

4. What is the chemical formula of rock 1f? (2) _____

5. The same elements used to form rock 1c are also main components in (2)
   a. seashells
   b. basalt
   c. granite
   d. pearls
   e. opal

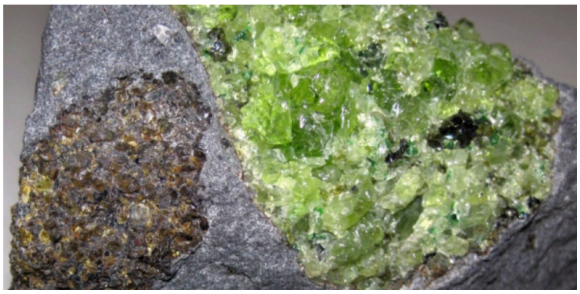
6. What gives 1d its blue color? (2)

_____

Section 5

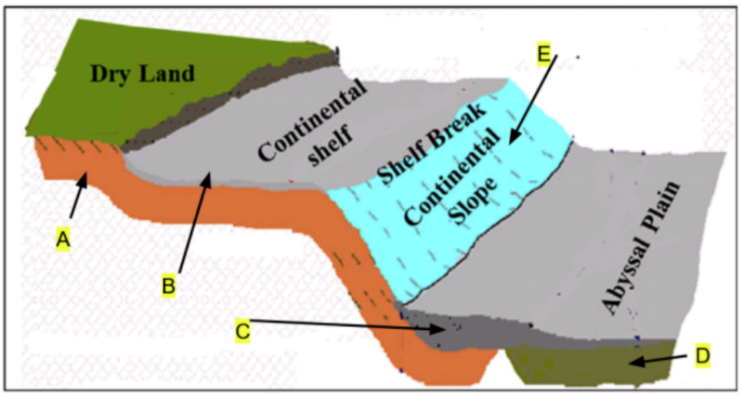
1. Identify the specimen on the right. (2)

_____



2. This specimen can be used to sequester _____. (3)

3. The gemstone variety of this specimen is referred to as _____. (3)



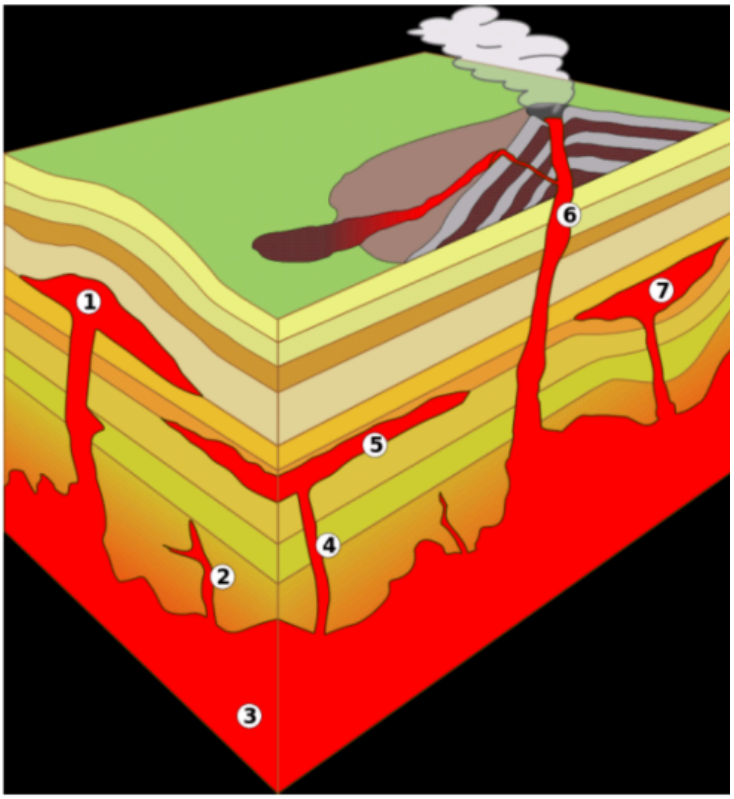
4. Using the letters in the diagram above, where would you most likely be able to find (4.5, 1.5 each) MULTIPLE ANSWERS MAY APPLY; ex: answer may be a and b

a. granite? _____

b. shale? _____

c. sandstone? _____

5. Label the diagram on the right. (7, 1 each)



1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

Section 6



1. Identify the rock/mineral shown on the right, including its Family. (2)

_____

2a. True or False? This rock/mineral is used in geochronological dating though Carbon-14 dating. (1)

_____

2b. If False then explain how it is correctly used in geochronological dating, if True leave it blank. (3)

_____

_____

3. Circle the correct characteristics of the rock/mineral. Select all that apply. (2)

a. Contain trace amounts of Uranium and Thorium
b. Exhibit a tabular habit
c. Is the rarest gemstone on Earth
d. Is a polymorph
e. Exhibit fluorescence

4. Describe the environment in which this rock/mineral is found in. (3)

_____

_____

5. Due to the uses of this rock/mineral, it is important that certain qualities are exellect as they affect the value of the rock/mineral. What are these characteristics and which institute established this standard? (5)

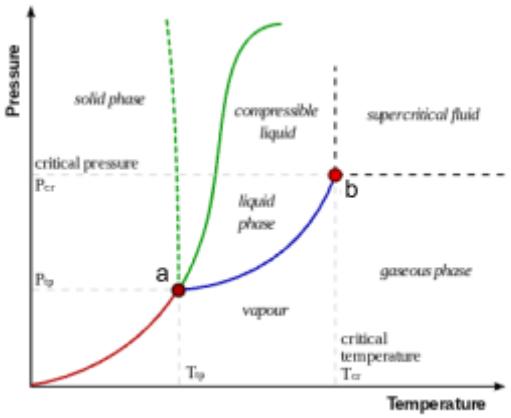
_____

_____

6. Label the points on the diagram below, given that it is a phase diagram of temperature/pressure. (2)

a. _____

b. _____



Section 7

a. 

b. 

c. 

d. 

e. 

f. 

1. Given that these are all rocks, identify the type of classification that applies to all the rocks. (1)
    a. Igneous
    b. Layered
    c. Sedimentary
    d. Metamorphic

2. Identify the rocks shown above and their streak color. (6, 0.5 each)

    A._____ _____, b._____ _____, c._____ _____
    d._____ _____, e._____ _____, f._____ _____

3. Explain the differences between a rock and a mineral. (2)

    _____

    _____

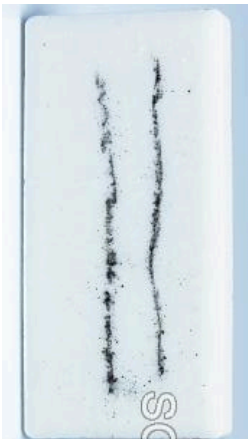
4. Match the rocks shown above with the correct environment. Some words may be used more than once, multiple numbers for one answer, or not at all. (6, 1 each)
    1. Batholiths, 2. Dikes, 3. Sills, 4. Laccoliths, 5. Volcanic necks, 6. Lava flows, 7. Pyroclastic flows

    A._____, b._____, c._____, d._____, e._____, f._____

5. Identify the correct order of grain size of the rocks shown above, from generally smallest to largest. (2)
    a. abcdef
    b. fbcead
    c. aebfcd
    d. edcbfa
    e. fabced

Section 8

While Erinna was taking a stroll she came across a rock/mineral, but was not sure which rock/mineral it was. Puzzled, she looked around and saw that a kind stranger had left some characteristics next to the specimen to help her identify it. The characteristic given were: Streak color - black, Density - 5.18 g/cm^3, Hardness - 5 to 6, Slightly soluble in HCl, and that it fractures rather than having cleavage.



1. Identify the rock/mineral detailed in the passage above. (2)
    a. Graphite
    b. Hematite
    c. Pyrite
    d. Magnetite

2. What notable characteristic of this rock/mineral is not noted above that could also be used to identify the specimen. (1)
    a. Smell
    b. Magnetism
    c. Piezioelectricity
    d. Fluorescence
    e. Taste

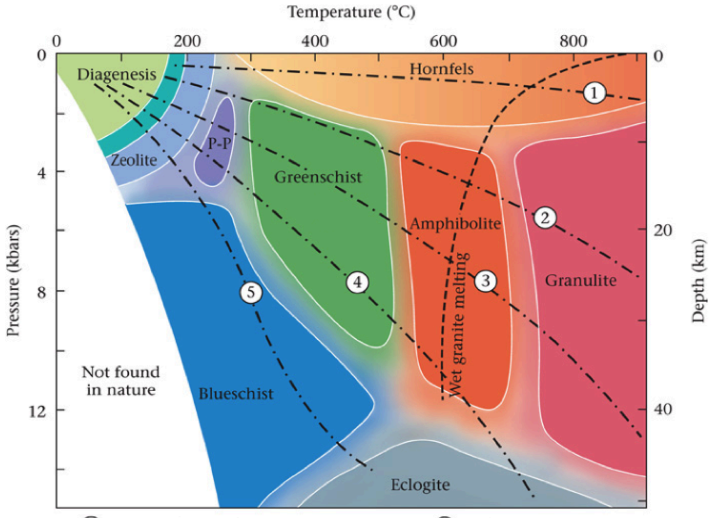
3. What is polymorphism in geology and how does it generally occur? (2)

    _____

4. Using the word bank below, fill in the blanks for the diagram below. (5, 1 each)
    Volcanic arc Stable continent Accretionary prism Contact metamorphism Collisional mountain belt

    1._____ 2._____ 3._____ 4._____ 5._____

5. Which of the following rocks is most likely to be found in the third section of the diagram shown to the right. (2)
    a. Gniess
    b. Marble
    c. Chalk
    d. Shale



6. How does the grade of metamorphism affect the formation of a rock and give an example of each. (4)

    _____

    _____

Section 9

1. What properties does the chemical composition of a mineral affect? Select all that apply (3)
    a. Luster
    b. Hardness
    c. Tenacity
    d. Solubility

2. Fill in the blanks in the statement below. (3)

    The _____ _____ Family known for their uncombined chemical _____ occur naturally and are typically more valuable for their industrial or jewelry uses.

3. Briefly explain why the the Silicate family is the largest group of minerals. (2)

    _____

    _____

4. Using the image shown to the right and the information provided, identify the rock/mineral. (2)
    - This specimen has cleavage, Streak color: white, and a Hardness: 2.

    _____



5. What is the specimens (4, 1 each)

    a. Luster? _____

    b. Density? _____

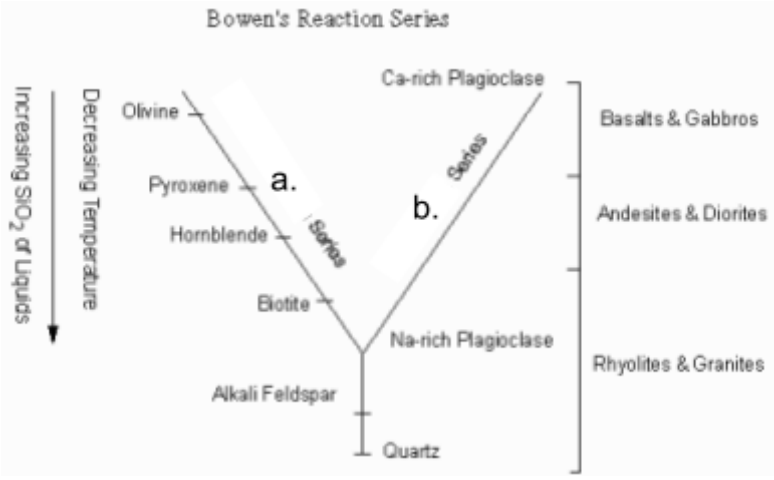
    c. Diaphaneity? _____

    d. Chemical formula? _____

6. Label the correct blanks about Bowen's Reaction Series shown to the right. (2)

    a. _____ Series

    b. _____ Series



Section 10

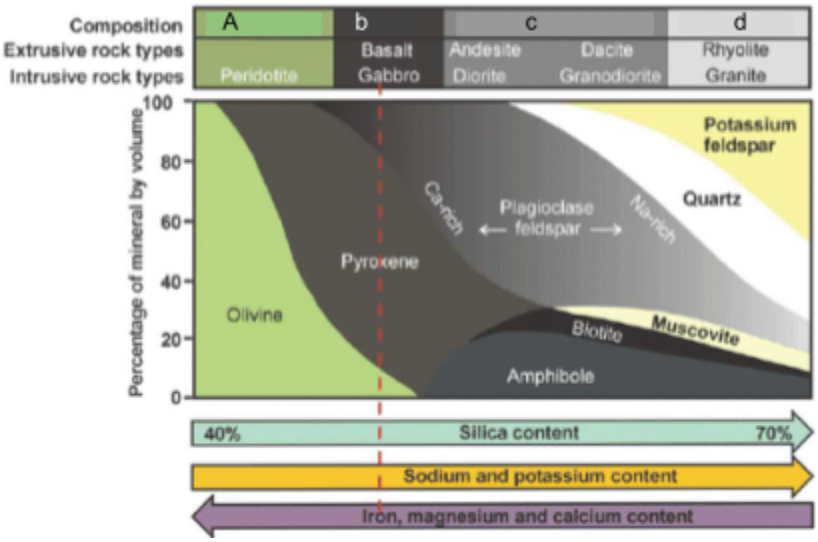
1. Label the diagram shown to the right with the correct words. (Hint: what categorization is used for the composition and essential minerals of this type of rock) (4, 1 each)

   a. _____

   b. _____

   c. _____

   d. _____



2. Which of the following rocks would most fit the description of having a high silica content, a high percentage of mineral by volume, and it being intrusive, using the image displayed above. (2)
   a. Dacite
   b. Peridotite
   c. Diorite
   d. Granite

3. Which of the following rocks/minerals would be most likely to be used for the manufacturing and technology industry? (2)
   a. Flourite
   b. Pumice
   c. Beryl
   d. Graphite

4. Identify the following rocks/minerals and their family/classification given the information provided below(EX. Schist, Metamorphic Rock). (8, 1 each)
   - Specimen a is formed in lacustrine environments
   - Specimen b has a density of 10.49 g/cm^3
   - Specimen c reacts with HCl by fizzing
   - Specimen d has a hardness of 1

a.  b.  c.  d. 

_____ _____ _____ _____

5. One of the specimen pictures above has a unique characteristic. This characteristic is double refraction. Identify which specimen above displays this characteristic and explain how it generally works. (3)

   _____

   _____